Forecasting Manhattan Citi Bike departures with public data

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Problem and implemented scope

This project forecasts served Citi Bike departures across Manhattan for each of the next three hourly intervals. At an origin T, it produces separate expected counts for T through T+1, T+1 through T+2, and T+2 through T+3. The intended contribution is a reproducible forecasting system that connects published trip outcomes, forecast weather, scheduled inference and delayed evaluation. Randy currently operates all workstreams using a local RAID and a private Databricks Free Edition workspace.

The original proposal targets individual stations and asks whether weather and subway disruptions improve forecasts. Public data supports a narrower initial experiment. An audit of the CityBikes archive found that its change-only observations disproportionately cover active stations and hours: only 0.147% of conservatively covered station hours have zero departures. Inferring station operation from those observations would condition evaluation on activity. Manhattan totals permit a useful benchmark while reliable operation histories remain unavailable. The target measures completed, served departures; unavailable bicycles and unobserved customer attempts prevent interpreting it as unrestricted demand. The model card records this scope decision.

Trip labels and data provenance

Published monthly Citi Bike trip files provide departure timestamps, station identifiers and coordinates. Full source archives remain outside the repository. A canonical manifest records discovery, schema profiling, source selection and published Parquet files. Derived exports read only the selected published-file ledger. This prevents duplicate extracted folders, superseded outputs or incomplete releases from entering a model through a directory wildcard. New releases are acquired separately from the original read-only archive and retain object metadata, retrieval time and SHA-256 checksums.

Normalization preserves audit fields and distinguishes departure eligibility from arrival eligibility. A missing destination does not automatically invalidate an otherwise usable departure. Modern files lack a source duration field, which remains null rather than being fabricated. Station identifiers remain strings. Documented normalization repairs truncated modern short-name identifiers, while duplicate spillover trips are removed only when an identical copy is already published by the release containing the start month. Per-source reconciliation requires profiled rows to equal published, quarantined and explicitly removed rows.

Each eligible departure is assigned geographically using its actual start coordinates and the downloaded Manhattan boundary. A complete borough-hour calendar includes zero geographic totals without asserting that any particular station operated. Timestamps are converted from New York time to UTC. Published naive timestamps cannot identify the two occurrences of a repeated autumn clock hour; those labels remain null, and affected forecast origins and profile exposure are excluded. Separate release-level aggregates preserve which monthly source supplied each count. The September ingestion report validates 4,982,855 new source records and 720 September borough hours containing 3,049,503 departures. Spillovers also revise 17 August geographic hours by 331 departures; original model metrics remain tied to their original data snapshot.

Weather and prospective transit evidence

The weather dataset uses individual ECMWF IFS initialization runs from the Open-Meteo Single Runs API, rather than observed weather substituted after the event. Requests identify one Manhattan reference point, four initialization cycles daily and a two-day forecast window. Returned grid coordinates, units, initialization and valid times remain in the normalized table. The weather datasheet describes the original benchmark coverage and the September extension, including unavailable requests and nullable values. A single reference point represents borough conditions, without establishing station microclimates.

Temporal alignment distinguishes instantaneous variables from accumulated precipitation. For a target interval 10:00–11:00, temperature and wind use the forecast valid at 10:00, while precipitation uses the preceding-hour accumulation ending at 11:00. Model-run initialization is different from public availability. Development assumes a seven-hour delivery delay, and eligible inputs must precede the origin by five minutes. Early archived runs include hindcasts and actual historical publication times are unverified. These conditions limit the strength of retrospective deployment claims even when feature timestamps pass consistency checks.

The running collector also saves public GBFS station information and MTA advisories. A dated mapping connects bicycle stations to subway reference points within 250 and 500 meters. Static GTFS supplies scheduled route, stop and direction associations. The official MTA alert dictionary distinguishes the subway alert namespace from bus namespaces; matching route numbers alone would create false associations. Within a selector, restrictions are combined with AND, while selectors combine with OR, following the GTFS specification. Advisory counts are prospective evidence and are absent from the deployed forecasting model.

Feature availability and model design

Predictors combine calendar cycles, horizon, released hour-of-week and hour-of-day departure profiles, and forecast temperature, squared temperature, precipitation, wind and a wet indicator. They exclude recent unavailable trip counts and properties of future trips, such as duration, destination or membership. Each profile uses monthly supplying-source cohorts available under a declared reconstruction rule: the source month ends, then 45 days elapse before assumed release at 00:00 UTC. Profiles retain both their source release and availability timestamp. Newly downloaded September outcomes therefore do not become retrospective June predictors.

The fixed candidate set contains pooled hour-of-day and hour-of-week arithmetic baselines, plus calendar and weather Poisson regressions with regularization strengths 0.1 and 1.0. Poisson regression produces nonnegative expected counts and allows a compact, interpretable comparison. Preprocessing fits on training data only. Selection minimizes validation mean absolute error, measured in departures per Manhattan hour. Missing or stale weather invokes a separate calendar model; zero placeholders are excluded from weather fitting and weather-complete comparisons. No calibrated predictive intervals are currently supplied.

The initial experiment trains from March 2025 through March 2026, validates during April–May 2026, and tests during June–August 2026. All three target intervals must remain inside one split. Weather-complete sets contain 28,149 training rows, 4,386 validation rows and 6,564 test rows. Repeated target hours across successive origins and three horizons make these rows dependent. Candidate comparisons use common keys, and row counts describe forecast cases rather than independent trials. The selected model remains weather Poisson with alpha 0.1, even though another candidate later has slightly better test error.

Retrospective results and release-aware evaluation

The initial selected model achieves test MAE 761.00, compared with 1,385.91 for the weekly profile and 815.81 for calendar Poisson. This is a 45.09% reduction relative to the weekly profile and 6.72% relative to the calendar model. Horizon-specific MAEs are 758.27, 760.55 and 764.19. The aggregate model can capture much of the predictable variation, while weather adds a smaller improvement over a fitted calendar model. These are development results under the stated archive and timing assumptions, as detailed in the saved evaluation.

A second analysis controls label availability during fitting and selection as well as feature construction. For each June, July and August origin month, it selects an already released validation month, fits candidates on earlier eligible labels, and refits the chosen candidate using only eligible released outcomes. Every fold records maximum supplying-source availability and the fit cutoff. The release-aware evaluation achieves pooled MAE 767.77 versus 1,386.49 for the weekly profile across 6,552 common rows. The smaller count reflects additional purging at monthly boundaries.

Monthly results reveal an important limitation hidden by a pooled headline. Selected weather MAE is 734.30 in June, 839.87 in July and 727.24 in August. In June, the calendar comparison performs better at 649.80 despite validation selecting weather. Thus weather does not improve every month, and model selection is uncertain across changing demand and weather conditions. The release-aware experiment reuses months already inspected during development; it strengthens the audit of the method but is not an independent untouched test. It does not promote or replace the active model.

Sensitivity checks apply hypothetical six-, seven- and eight-hour weather delays to the frozen model without refitting. On common validation keys, MAE is 557.89, 558.63 and 562.78 respectively. Similar errors across assumptions do not verify provider publication times. More broadly, the original May validation labels would not have been released at the first June origin under the 45-day rule. The original experiment therefore supports retrospective model development; the monthly analysis addresses that particular fitting limitation while retaining the weather archive caveat.

Deployment and application demonstration

An October 20 extension evaluates bounded ensembles on the conditional 32-station cohort. June validation selects weather histogram boosting ahead of random forest and Poisson. On 58,965 inspected September examples, MAE is 2.1338 departures per station-hour for weather boosting, 2.8964 for weather forest and 3.4803 for weather Poisson. The Poisson reference reproduces the earlier pilot, while the boosting result shows that nonlinear station/calendar/weather interactions can materially improve this restricted study. September had already been inspected, so later confirmation is still required before promotion.

The availability extension compares the existing boosting counts with persistence and two forests, selecting by outcome/horizon on August validation. No forest wins. Persistence remains selected for bike counts at 15 and 30 minutes and dock counts at 15 minutes; 31-leaf boosting serves the remaining horizons. Cloud execution reproduces local results and stays below 1.2 GB peak RSS. These development runs write versioned volume and MLflow artifacts without changing active models, stable tables or schedules. Full metrics and protocols are in the ensemble report.

This measured scale does not support migrating the model layer to Spark ML. The station and availability packages contain 234,238 and 492,270 rows, fit within roughly 1.2 GB of process memory and complete within the existing budget. Spark would add scheduling, serialization and estimator-parity work while the

material project constraints remain source coverage, release timing and later-period confirmation. Databricks is still useful for managed execution and MLflow; it does not require a distributed estimator. If future minute-level, multi-year or multi-city data exceeds the bounded workflow, Spark should first be benchmarked for cloud-resident ingestion, joins, chronological window features and incremental partition refreshes. The existing package contract can then continue feeding validated scikit-learn models until training itself exceeds single-node capacity. The Spark scaling decision records the adoption triggers and parity requirements.

Bulk normalization and geographic scans run locally with bounded DuckDB memory and sequential archive reads. Compact Parquet tables, immutable model artifacts and helper code are uploaded to versioned Unity Catalog volume directories. Databricks checks hashes and counts, then replaces stable current Delta tables whose rows identify the immutable package in `dataset_version`; source packages remain on the RAID and in volumes. Training evidence is recorded in MLflow. Serving pins a compatible runtime and scikit-learn version; saved and reloaded predictions agree with training outputs. This separation keeps large trip processing outside the constrained cloud account while preserving reproducible model inputs.

The hourly job runs at minute 55 UTC, collects public sources, then scores the next top-of-hour origin. It persists three predictions with model version, input provenance, fallback reason and actual publication timing. Forecast keys preserve the first prediction during retries. A deliberately late retry exercised stale-weather fallback while retaining the original forecasts and timestamps. One concurrent run, bounded timeouts and explicit failure reporting reduce overlapping writes. The deployed model remains frozen; release collection and outcome updates do not automatically retrain it.

A separate daily evidence job watches public trip releases, imports only validated prepared outcomes, and normalizes accumulated station and advisory snapshots. Another daily job checks forecast delivery and source freshness. Its health status describes operational delivery rather than predictive accuracy. The dashboard displays forecast intervals, publication timing, model comparisons, mapping coverage and pending evaluation. The demo script pairs three slides with a five-minute navigation sequence and saved artifacts for an interrupted session.

Reproducibility and maintenance

The repository separates code, documentation and small validation summaries from full datasets on the RAID. Immutable manifests tie each label, feature and portable handoff to input hashes and row counts. New prepared outcomes pass a local staging gate that checks complete hourly calendars, unique keys, nonnegative counts, repeated-clock nulls and selected-source provenance. Cloud import repeats the essential checks and publishes a ready receipt only after the required files exist. This design allows a reviewer to trace an outcome update independently of the forecasting model.

Prospective derivations use append-only raw tables and completed-batch publication receipts. Failed partial writes remain outside public views, and replay deduplicates logical keys within a processing version. The first zero-input replay preserved published counts. Tests cover temporal boundaries, station identifiers, duplicate removal, missing sources and advisory resolution; cloud checks exercise runtime behavior and persistence. Free Edition availability and public endpoints can still interrupt collection. The runbook therefore records job schedules, separate pause controls, package locations and recovery steps. A saved PDF deck and dated validation summaries provide a demonstration fallback without presenting stale snapshots as current results.

Limitations and next steps

Production accuracy requires genuine forecasts issued before their targets and subsequently matched to published departures. September historical outcomes cannot evaluate live forecasts first issued in October. Evaluation therefore remains pending for those forecasts until corresponding monthly outcomes mature. A release listing or provider modification timestamp does not establish when data first became publicly usable. The watcher records first observations prospectively, and the ingestion gate checks archive completeness before any prepared outcome becomes available to evaluation.

The first normalized prospective batch contains 47,858 station observations, 9,078 advisory observations and 95,716 station/radius feature rows. Seven observed station configuration changes were capacity updates. These sampled observations support a growing audit trail, without reconstructing historical operation or exact change times. Stale feeds, missing stations and incompatible mappings yield unknown values. Zero matched advisories does not establish normal subway service. Estimating alert effects still requires adequate historical coverage, a defensible station population and an evaluation design that addresses confounding.

The next substantive milestones are class-workspace deployment, accessible publication of the documented weather contribution, and a sustained production observation period. The portable weather package is ready for transfer, but a private import does not satisfy cross-team access. Historical station operation and geography remain necessary before returning to station-level predictions. This project shows that the current public inputs support an operational borough benchmark, with explicit records of what was available, how predictions were preserved and which claims the resulting evidence can sustain.